

COVERAGE ACROSS THE HORIZON

Horizon- and Channel-Conditional Adaptive Conformal Calibration for Long-Term Time-Series Forecasting

Original research gap identified and formulated by **Aarush Pandit** - B.Tech (Information Technology), VIT Vellore
Research Concept Note - **Idea Lock v1.0** - 16 July 2026 - Literature re-verified through mid-July 2026 - Monochrome working document

PART I - THE IDEA AND THE ORIGINAL GAP

1. Thesis

The three papers that define modern long-term time-series forecasting (LTSF) - Informer, DLinear, and PatchTST - disagree about almost everything except their evaluation contract: deterministic point forecasts scored by MSE/MAE on a shared benchmark suite at horizons of 96-720 steps. None of the three emits any measure of predictive uncertainty. This document locks a project that (i) audits, systematically and for the first time, how badly and precisely *where* these field-defining backbones are miscalibrated when standard interval methods are attached; (ii) fixes it with a lightweight, post-hoc, distribution-free calibration layer whose target coverage is conditioned on horizon-bucket and channel and adapted online; (iii) quantifies the marginal-versus-joint (whole-path) coverage trade-off at horizons up to 720; and (iv) evaluates calibrated intervals at the level of *decisions*, on real building-energy data under temporal distribution shift. No new forecaster is trained; the entire study runs on free-tier compute.

2. The anchor triad (verified primary sources)

The triad is deliberately adversarial: P2 attacks P1's central claim on P1's own benchmark, and P3 answers P2 on the identical protocol. Because the three disagree on architecture yet share one blind spot, that blind spot is a property of the field's evaluation contract rather than of any single paper - which is what makes the gap in Section 3 defensible as a finding.

Paper	Venue / Year	Role in the dialectic	Link
P1 - Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. Zhou, Zhang, Peng, Zhang, Li, Xiong, Zhang.	AAAI 2021, vol. 35, pp. 11106-11115 (Best Paper)	Thesis: long horizons need high-capacity attention made efficient (ProbSparse attention, distilling, one-shot decoding). Established the benchmark protocol.	arxiv.org/abs/2012.07436
P2 - Are Transformers Effective for Time Series Forecasting? (DLinear). Zeng, Chen, Zhang, Xu.	AAAI 2023, vol. 37(9), pp. 11121-11128	Antithesis: one-layer linear maps beat all prior Transformers on nine datasets; gains attributed to direct multi-step decoding, not attention.	arxiv.org/abs/2205.13504 doi:10.1609/aaai.v37i9.2631Z
P3 - A Time Series is Worth 64 Words: Long-term Forecasting with Transformers (PatchTST). Nie, Nguyen, Sinthong, Kalagnanam.	ICLR 2023	Synthesis: Transformers win again via patch tokenization plus channel independence (reported approx. 21% overall MSE reduction vs. best prior Transformer results).	arxiv.org/abs/2211.14730

3. The gap (the original finding)

G1. All three anchors optimize, decode, and report exclusively deterministic point forecasts under MSE/MAE: no intervals, no quantiles, no notion of coverage - at exactly the horizons where point forecasts are least trustworthy and where the motivating applications (electricity planning, traffic, epidemics) are risk-sensitive.

G2. This is a structural divide, not an accident of three papers: the ProbTS benchmark (NeurIPS 2024 Datasets & Benchmarks, arXiv:2310.07446) documents that the literature split into long-horizon *point* forecasting and short-horizon *probabilistic* estimation, with neither line serving both needs.

G3. Work from 2024-2026 is converging on this hole from the conformal-prediction side (Section 4), but as of the lock date no work combines the four wedge components below on the canonical LTSF suite with these backbones. A June 2026 preprint (arXiv:2606.09473) independently argues that training-free conformal intervals are a *mandatory but currently missing baseline* in probabilistic forecasting evaluation - external confirmation that the hole is recognised and unfilled.

CLAIMED WEDGE - bounded statement of originality (as of 16 July 2026)

No published work provides, **in combination: (W1)** a systematic conditional-coverage audit of the canonical LTSF suite (ETT x4, Electricity, Traffic, Weather, ILI) at horizons 96-720 across Informer, DLinear, and PatchTST; **(W2)** a post-hoc, distribution-free calibration layer with horizon-bucket x channel conditioning and online (ACI/PID-family) adaptation at that scale; **(W3)** an explicit marginal-versus-joint (whole-path) coverage analysis at H up to 720; and **(W4)** a decision-level evaluation of calibrated intervals on real building-energy data under temporal shift. **Explicitly NOT claimed as novel:** split conformal prediction; per-horizon split conformal (MSCP - an established baseline, see F1); online adaptive conformal in general. The contribution is the audit, the conditional-adaptive combination, the joint-coverage analysis, and the decision utility, at LTSF scale, under full protocol hygiene. The wedge survives only while unoccupied: re-verify monthly (Section 12).

4. Prior-art fence - nearest works, each verified, and why none closes the wedge

Work	Reference	Why it does not close the wedge
ProbTS (Zhang et al., NeurIPS 2024 D&B)	arxiv.org/abs/2310.07446	Documents the point-vs-distributional divide; a benchmark of trained probabilistic models, not a post-hoc calibration method.
Adaptive Conformal Inference (Gibbs & Candes, NeurIPS 2021, 34:1660-1672)	NeurIPS 2021 proceedings	Foundational online conformal update; one-dimensional target stream, no horizon/channel structure.
Conformal PID Control (Angelopoulos, Candes, Tibshirani, NeurIPS 2023)	arxiv.org/abs/2307.16895	Online adaptive intervals; scalar series and short horizons (4-week COVID, demand, returns, temperature) - not the LTSF 96-720 grid.
MSCP: per-horizon split conformal (Wang & Hyndman line, 2024/2026)	arxiv.org/abs/2410.13115	Now a documented baseline paradigm (horizon-specific scores/quantiles). Treated here as a baseline , never as the contribution (fix F1).
AcMCP + multi-step PID (Wang & Hyndman, updated Feb 2026)	arxiv.org/abs/2410.13115	Handles autocorrelation of multi-step errors with long-run guarantees; demonstrated at small H (approx. 3) with ARIMA/ETS/NN - no LTSF backbones, no channel conditioning, no decision layer.
Bellman Conformal Inference (Yang, Candes, Lei, 2024)	arxiv.org/abs/2402.05203	Joint multi-step control via dynamic programming; reported to ignore cross-horizon temporal dependence and to be computationally heavy per step.
CopulaCPTS (Sun & Yu) and Schlembach et al. (COPA 2022; ext. 2025)	arxiv.org/abs/2212.03281	Multivariate multi-step conformal; small horizons, fixed calibration; not LTSF-scale backbones. CopulaCPTS venue of record to be re-confirmed.
O2CP: optimization-based online multi-step CP (2025)	arxiv.org/abs/2508.13362	Two-layer online CP across time and horizons; its own conclusion names decision-coupled calibration as open future work - direct evidence W4 is unoccupied.
Dual-Splitting CP for multi-step forecasting (2025)	arxiv.org/abs/2503.21251	Multi-step CP variant; not the LTSF regime, backbones, or conditional-coverage focus.
PatchTST + ACI for PV (Energies, 2025)	mdpi.com/1996-1073/18/18/5000	Hour-ahead, single 5 kW rooftop site; confirms applied appetite, is not a systematic LTSF audit.
Incident-aware ST-Transformer + ACP for traffic (2026)	arxiv.org/abs/2603.16857	Domain-specific multi-horizon calibration at 1-4 hour horizons on state DOT data; not the canonical suite or backbones.
DeRegIME: regime mixtures under shift (2026)	arxiv.org/abs/2605.19231	Nearest territorial neighbour: probabilistic forecasting on the LTSF suite with DLinear/PatchTST backbones - but a trained parametric density model, not distribution-free post-hoc calibration with coverage control. Differentiate explicitly in the paper.
Report the Floor (June 2026)	arxiv.org/abs/2606.09473	Argues training-free conformal intervals are a mandatory, currently missing baseline - supports the audit spine and the timing.
Conformal Seasonal Pools (2026)	arxiv.org/abs/2605.03789	Training-free probabilistic forecasting via seasonal pooling; not conditional-coverage on LTSF backbones.
CP-for-forecasting benchmark survey (Jan 2026)	arxiv.org/abs/2601.18509	Closest benchmarking work; formalises MSCP and online families. Read first (Stage S0) , verify dataset/model overlap, position against it explicitly.

Scope fence (architecture, deliberately out of scope): cross-channel and exogenous-variable modelling belong to iTransformer (ICLR 2024, arxiv.org/abs/2310.06625) and TimeXer (NeurIPS 2024, arxiv.org/abs/2402.19072); this project contributes calibration, not architecture.

5. Recheck log - gaps found on re-verification (16 July 2026) and fixes locked into v1.0

ID	Gap found on recheck	Fix locked
F1	Per-horizon split conformal is now a named baseline paradigm (MSCP), so it cannot be the novelty claim.	Novelty boundary redrawn: MSCP demoted to baseline; the claim is the W1-W4 bundle in the wedge box.
F2	Multi-step online CP methods with horizon structure exist (AcMCP, MPID, Bellman CI, O2CP).	All four added as named comparisons/baselines; differentiation stated per method in the fence table.

ID	Gap found on recheck	Fix locked
F3	The standard LTSF test protocol uses stride-1 sliding windows, so residuals of adjacent windows are heavily dependent; naive calibration is optimistically biased.	Calibration uses strided/non-overlapping windows; dependence-aware AcMCP is a comparison, not an oversight.
F4	Fitting 720 independent horizon quantiles on a finite calibration block explodes variance.	Bucketed conditioning: K log-spaced horizon buckets, K in {4, 6, 8, 10}; K is an ablation.
F5	Exact finite-sample marginal coverage does not survive non-exchangeability; exact distribution-free conditional coverage is impossible in general.	Guarantee honesty: claim only ACI-family long-run adaptive coverage plus bucketed (approximate) conditioning; a Theory-Scope box goes in the paper.
F6	Per-cell 90% coverage over an H x C grid is not 90% whole-path coverage; conflating them is a desk-reject risk.	Both reported; joint coverage via max-normalized-score, Bonferroni, and copula variants, with the width cost of each guarantee quantified.
F7	Known LTSF evaluation pitfalls (e.g., drop-last batch truncation inflating test scores in common codebases) can silently corrupt coverage numbers.	Protocol hygiene rules in Section 10: drop_last disabled, sequential splits, fixed seeds, exact protocol published with code.
F8	Quantile-head (pinball-trained) baselines can be unfairly weak if under-tuned, inviting reviewer pushback.	Equal tuning budget rule for all trained baselines; Chronos zero-shot quantiles included only as an uncalibrated reference.

6. Contribution claims (what the paper will assert)

C1 - Audit. The first conditional-coverage map of the canonical LTSF suite across the three field-defining backbones: where, along horizon, channel, and regime, do standard interval methods fail? **C2 - Method.** A bucketed horizon x channel conditional adaptive conformal layer: post-hoc, model-agnostic (demonstrated by backbone swap), $O(1)$ per step, NumPy-implementable. **C3 - Analysis.** The marginal-versus-joint coverage trade-off at H up to 720, with the width price of each guarantee. **C4 - Utility.** The decision-level value of calibrated intervals on real building-energy meters under a chronological 2016-to-2017 shift. Honest expectation: the median outcome for a solo undergraduate execution is a workshop paper plus arXiv, with a realistic path to a mid-tier journal via the energy framing; a top-venue main-track acceptance would be an outlier.

PART II - IMPLEMENTATION STRATEGY

7. Pipeline overview

frozen backbone -> residual tensor ($H \times C \times t$) -> scale-normalized scores -> per-(bucket $\times$ channel)
adaptive quantile -> intervals -> coverage / width / decision evaluation

Each backbone is trained once per dataset and frozen. Rolling calibration windows produce residuals; nonconformity is the residual normalized by a rolling per-channel scale estimate (median absolute deviation vs. standard deviation is an ablation). One adaptive quantile tracker is maintained per (horizon-bucket, channel) cell using an ACI/PID-style update toward the target level. Intervals, marginal and joint coverage, width, and decision metrics are computed downstream. Everything after the backbone is plain NumPy.

8. Stage plan with exit criteria

Stage / window	Objective	Key actions	Exit criterion
S0 mid-late Aug 2026	Positioning	Read arXiv:2601.18509, 2410.13115, 2307.16895, 2508.13362 closely; write a 1-page positioning note; confirm no exact wedge collision.	Wedge statement survives unchanged, or is amended in writing.
S1 Sep, weeks 1-3	Reproduction	DLinear + PatchTST on ETTh1, ETTm2, Weather at H in {96, 336, 720}; archive full per-horizon, per-channel residual tensors (calibration + test).	Point MSE within about 5% of published values; residual tensors archived.
S2 late Sep - early Oct	Audit + DECISION GATE	Global split conformal vs. per-horizon MSCP; coverage-vs-width plots across the horizon grid; per-channel breakdown.	Gate resolved: method-led vs. audit-led spine (see note below).
S3 Oct	Method	Bucketed horizon $\times$ channel conditioning + per-cell ACI/PID adaptation; joint coverage via max-score / Bonferroni / copula; add Informer as third backbone.	Conditional-adaptive layer beats global and MSCP baselines on worst-cell coverage at comparable width, or the audit spine is adopted.
S4 early Nov	Shift & robustness	Rolling coverage traces through nonstationary segments (ILI, Exchange-like); full ablation grid.	Ablation table complete; adaptation shown to matter (or not) under shift.
S5 mid Nov - early Dec	Energy case study	BDG2, about 50 meters, chronological 2016-to-2017 split; decision task: interval-gated peak-demand flagging (fallback: pre-cool trigger); report decision cost/regret alongside coverage.	Decision-level result exists in either direction; negative results reported honestly.
S6 Dec 2026	Write & release	Paper draft, code release, arXiv submission; then workshop CFPs Jan-Feb 2027.	arXiv preprint live with code; two target CFPs identified with dates verified.

S2 DECISION GATE - the coverage-vs-width plot

Hold out a calibration block. Compute one global conformal quantile and per-horizon (MSCP) quantiles from the same residuals. On the test block, plot empirical coverage and normalized interval width against horizon step (1 to 720) for both. **If** global coverage sags below target at long horizons while per-horizon/conditional calibration restores it, the residual scale is horizon-heterogeneous and the paper is **method-led**. **If** global coverage is already flat at target, conditioning is cosmetic on periodic benchmarks and the paper pivots to **audit-led**: charting where calibration breaks (nonstationary segments, BDG2 shift) - a framing the June 2026 mandatory-baseline argument (arXiv:2606.09473) independently supports. Both branches are publishable; the gate sets emphasis, not viability.

9. Datasets, code, and links (all public; all resolved 16 July 2026)

Asset	Link	Notes
LTSF benchmark suite (ETTh1/2, ETTm1/2, Electricity, Traffic, Weather, ILI)	github.com/yuqinie98/PatchTST github.com/thuml/Time-Series-Library	Official PatchTST repository and the Time-Series-Library provide data download scripts, reference implementations of all three backbones, and the standard split protocol.
Building Data Genome 2 (BDG2) - energy case study	github.com/buds-lab/building-data-genome-project-2 doi:10.1038/s41597-020-00712-x	3,053 hourly meters, 1,636 non-residential buildings, 2016-2017 (Miller et al., Scientific Data 7:368, 2020). Includes the ASHRAE GEPIII competition subset. Use about 50 electricity meters; chronological 2016-to-2017 split induces genuine shift.
Anchor and fence papers	Sections 2 and 4 above	All arXiv identifiers in this document were resolved live on the lock date.

10. Comparison set, metrics, ablations, protocol hygiene

Backbones (frozen point forecasters): DLinear, PatchTST, Informer - the anchor triad, via their reference implementations.

Interval methods compared: Gaussian residual intervals; pinball-trained quantile heads (equal tuning budget); global split conformal; per-horizon split conformal (MSCP); ACI; Conformal PID; CopulaCPTS; AcMCP/MPID where code permits (else reimplemented or explicitly scoped out in writing); proposed conditional-adaptive layer; Chronos zero-shot quantiles

(arXiv:2403.07815) as an uncalibrated reference only.

Metrics: coverage error (mean and worst cell over the horizon-bucket x channel grid), rolling coverage traces, normalized interval width, Winkler/interval score, quantile loss, joint whole-path coverage, and wall-clock overhead of the calibration layer. **Ablations:** conditioning off / horizon-only / channel-only / both; bucket count K; adaptation step size; calibration window length; scale normalization (rolling MAD vs. std); strided vs. overlapping calibration windows; backbone swap (the model-agnosticism claim).

Protocol hygiene (binding rules): (1) sequential splits only, never random; (2) drop-last batch truncation disabled everywhere; (3) strided calibration windows to control residual dependence (F3); (4) equal tuning budget for all trained baselines (F8); (5) fixed and published seeds, three seeds minimum for trained components; (6) worst-cell coverage reported alongside means - averages hide exactly the failures this project is about; (7) exact protocol and code released with the paper.

11. Timeline, effort, and compute budget

Placement-season aware: the first half of August 2026 stays protected for DSA and aptitude preparation (deadline 5 August); research ramps from mid-August at roughly 8-10 focused hours per week, with stage windows as in Section 8 and slack absorbed by the Dec writing block. Compute: DLinear trains on CPU in minutes; PatchTST needs a few T4 GPU-hours per dataset; Informer is comparable. The full matrix fits inside Kaggle's free GPU quota (about 30 GPU-hours/week - verify current terms) with Colab free tier as overflow; the calibration layer itself is NumPy and effectively free. Storage: residual tensors are a few hundred MB total.

12. Risks and kill-switches

Risk	Trigger	Response
R1 Scooped mid-project (the space is moving monthly)	A new preprint occupies W1-W2.	Post to arXiv immediately after S3 rather than waiting for S5; shift weight to W3-W4 and the audit spine; standing monthly scan (Section 13).
R2 Global conformal already flat at target everywhere	S2 gate output.	Pivot to audit-led spine: where and when calibration breaks (nonstationary segments, BDG2 shift). Publishable per the mandatory-baseline argument.
R3 BDG2 messiness sinks the case study	Excess missingness/outliers in chosen meters.	Fallback decision task on the Electricity benchmark only; BDG2 demoted to appendix.
R4 Reviewer flags guarantee overclaim	Any claim of finite-sample or exact conditional coverage.	Pre-empted by F5: Theory-Scope box; only long-run adaptive coverage plus approximate bucketed conditioning is ever claimed.
R5 Compute or time overrun	Stage slips by more than 3 weeks.	Drop Informer (keep DLinear + PatchTST); MVP = S1 + S2 + S3 on four datasets = a workshop paper on its own.
R6 Placement-season squeeze	Interview load spikes (TCS NQT window, drives).	Same MVP contraction as R5; S5 energy study becomes v2 / journal extension.

13. Publication ladder, patent position, and the lock

Publication ladder: arXiv on completion of S6; first external targets are time-series or climate/energy workshops at ICLR/ICML/NeurIPS and the AAAI student track (CFP dates to be verified at S6 - not verified here); application venues where the framing lands well: ACM e-Energy or BuildSys short paper; journals: Applied Energy, Energy and Buildings, or Engineering Applications of AI, with the International Journal of Forecasting as the stretch target.

Patent position (India), locked: under Section 3(k) of the Patents Act, an algorithm or a computer programme per se is excluded; the calibration layer claimed alone is publishable but not patentable. The technical-effect route (the Delhi High Court's Ferid Allani line and CRI examination practice) would require embodiment in a concrete control system - e.g., an interval-gated HVAC pre-cooling or dispatch controller with a demonstrated technical improvement - where prior art from Daikin, Honeywell, and Siemens is dense and inventive step is doubtful. Decision: treat publication as the defensive disclosure; no filing spend now; revisit only if a physical control-loop embodiment is built, with a professional prior-art search. Before any HVAC-control build, re-read the Daikin internship agreement for surviving IP and confidentiality clauses, and keep the project strictly on public data. This is an assessment, not legal advice.

IDEA LOCK AND CHANGE CONTROL

Scope is frozen as of v1.0, 16 July 2026. Any scope change requires a written delta note against Sections 3-5 stating what changed in the literature or the evidence. Standing rule: on the first Monday of each month, re-run the wedge-collision scan (arXiv: conformal prediction + long-term forecasting; probabilistic LTSF; new citations of the fence works) and log the result in one line. If a new work occupies W1-W2, shift weight to W3-W4 and the audit spine. If all four wedge components are occupied, this document is superseded and a fresh gap search begins - do not sunk-cost a dead wedge.

All arXiv identifiers and repository links in this document were resolved live on 16 July 2026. Statements attributed to papers reflect their published abstracts and reported claims; assessments of limitations, novelty boundaries, and risk are the author's own working judgements. Prepared as a private working note; not a manuscript.